Marwan Emira

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Education

American University of Sharjah *August 2022 – June 2026*
BSc in Electrical Engineering CGPA: 3.74

- Coursework:

1- RF Transceiver Design & Analysis (Grade: A), VLSI Design (Grade: A-), Introduction to Radar Systems (Grade: B+)

2- Currently enrolled in Master Micro (gm/Id Methodology), the self-paced online course on analog IC Design

- Awarded two Undergraduate Research Grants for timely research efforts in Communications Engineering
- Published IEEE conference paper on RF power amplifier modeling (RIVF 2025)

Experience

Visiting Researcher (Part-time) *Dubai*
MIT Senseable City Lab *August 2026 - Present*

- Currently involved in two global projects across MIT Senseable City Lab's urban development research community

NCC and Crew Scheduling Intern *Dubai*
flydubai *July 2026 - September 2026*

- Observed Network Control Centre (NCC) operations, including interaction log review during incident handling, application of ZED buffer, FDP (flight duty period), acclimatized-crew rules, and delay code correction workflows from airport-level entry
- Contributed to case studies, risk assessments, and Irregular Operations (IROPS) reports through contingency analysis of past and potential operational disruptions, referencing delay analysis data (TCO/RAM)
- Simulated entire maintenance workflows and OEM design reviews with the engineering & maintenance control teams

Asset Management Intern *Sheikh Zayed City, Cairo*
ADES — Advanced Energy Systems *July 2026*

- Shadowed troubleshooting, commissioning, and repair/inspection activities on top drive systems and associated drive equipment, gaining exposure to preventive and corrective maintenance procedures
- Reviewed and interpreted electrical one-line diagrams for rig power distribution systems (generators, switchgear, SCR/VFD houses, top drives) to build familiarity with fault isolation and system architecture
- Conducted root cause analysis on drawworks failures caused by Profibus communication losses between PLC and VFD systems, then coordinated with peers to build and present technical findings to senior management
- Used Oracle EAM to log, track, and close out work orders for electrical maintenance activities, supporting maintenance planning across the mixed offshore/onshore rig fleet

Undergraduate Research Assistant *University City, Sharjah*
American University of Sharjah *August 2024 - May 2026*

- Performed RF measurement and benchmarking using laboratory instrumentation and connectorized transmitter chains
- Implemented and evaluated ML-based models for RF system characterization, transmitter linearization and benchmarking
- Developed compact technical documentation detailing methodologies, data pipelines, and system architectures supporting research continuity
- Secured $18,000+ in research funding by authoring winning proposals across two consecutive funding cycles.

Communications Systems Engineer (Software-Defined Radio) *August 2025 – April 2026*
IEEE ComSoc

- Led a technical workshop instructing students in the design and implementation of signal-processing flowgraphs using GNU Radio, emphasizing practical, system-level thinking
- Demonstrated end-to-end development workflows by guiding participants in the automatic generation, inspection, and manual refinement of Python scripts derived from GNU Radio flowgraphs
- Introduced students to large-scale, commercially inspired communication systems, highlighting modular design, component interoperability, and system-level complexity aligned with commercial communications systems

Reporting Task Force (Volunteering) *COP28, Dubai*
Ministry of Education *December 2023*

- Spearheaded the reporting and communication efforts for COP28, resulting in a 95% improvement in the efficiency of delivering high-level summaries and insights from sessions to the Ministry of Education
- Synthesized complex technical and policy discussions into concise reports for senior stakeholders

PCB Fabrication and Development *2023-2026*

Received a Professional Certificate from AUS & Mahara-Tech

- Designed a Microwave Class A Power Amplifier PCB using standard EMC-compliant techniques
- Executed PCB fabrication through chemical processes, including multi-layer PCB development

Projects

(Capstone) SYSTEM DESIGN & AUTOMATION – ADRV9026 Evaluation Board Setup *Completed Dec 2025*

Received the Undergraduate Research Grant from the American University of Sharjah (AUS)

- Documented a complete ADRV9026+ADS9V2EBZ-based transceiver evaluation system, enabling reproducible RF characterization, benchmarking, and research in *5G* signal analysis
- Implemented multi-chip synchronization over the ADRV9026's JESD204B/C serial interface to align LMFC boundaries across converters, achieving precise timing alignment for DUT characterization and MIMO experimentation
- Developed embedded control scripts (C, C#, and IronPython) to automate configuration loading, waveform generation, and transceiver calibration through the TES software and .NET APIs

LINEARIZATION OF POWER AMPLIFIERS (PAs) *Archive*

Received the Undergraduate Research Grant from the American University of Sharjah (AUS)

- Applied state-space ML-based behavioral model architectures with memory (BiLSTM, ARVTDNN) for PA linearization
- Implemented & validated novel techniques for matching non-linearities of PAs with the same architecture
- Developed frequency-based novel cost functions with relevant performance indicators that decreased model training time by 90% with for the same performance
- Collaborated with a team to develop a DPD System on Chip (SoC); verified operation in the 5G NR band

RF HOMODYNE TRANSMITTER DESIGN *April 2025*

Designed a DCS-1800 band Homodyne Transmitter Lineup for the RF Transceiver Design & Analysis course

- Designed and simulated a superheterodyne RF transmitter converting a 100 MHz digital IF signal to a 1.84 GHz RF output within the DCS-1800 band using Mini-Circuits components
- Modeled amplifiers, mixers, and filters in ADS using datasheet-derived S-parameters and nonlinear characteristics; evaluated system behavior using harmonic balance analysis

PLL FOR GROUND PENETRATING RADAR *January 2025*

Designed a Phase Locked Loop (PLL) system for a Ground Penetrating Radar transmitter

- Validated the design using ADIsimPLL software and Analog Devices' ADF4159 frequency synthesizer
- Achieved exceptionally low frequency and phase errors, optimizing chirp generation and signal integrity

Competition Entry — AutoTarget Challenge: Smart Ball Redirection Robotics *April 2026*

IEEE UAE Student Day 2025–26

- Implemented a 6-state Kalman filter (position, velocity, acceleration) in Python (NumPy) for robust ball trajectory estimation, maintaining prediction continuity during occlusion frames
- Designed an HSV color-masking pipeline in OpenCV operating at 320×240 resolution, outperforming ML-based detectors for this latency-critical embedded target

VLSI Full Adder Design — Schematic to Layout Flow (22nm) *May 2026*

Designed a simple adder on 22nm technology.

- Designed, sized, and simulated a NAND-based 1-bit CMOS full adder from gate-level schematic through physical layout, using LTspice for transient analysis and Electric VLSI for standard-cell layout and parasitic extraction.
- Built end-to-end EDA fluency across schematic capture, SPICE simulation, physical layout, and parasitic sign-off — a workflow subsequently applied to a CIC-based beamformer design (180nm).

Publications

Dec 2025: M. Emira, M. Ahmed, and O. Hammi, "Reduced-Complexity Transfer Learning Based BiLSTM Network for Multi-Amplifier Behavioral Modelling," presented at the 19th Int. Conf. on Computing and Communication Technologies (RIVF 2025), Ho Chi Minh City, Vietnam, Dec. 20, 2025, Paper no. 4.

Technologies

RF & Simulation/General Tools: ADS, PathWave (SystemVue), MATLAB, KiCad, SPICE, Multisim/Ultiboard
Software & Platforms: .NET, Visual Studio, VS Code, Excel/Word (Microsoft 360)
Programming Languages: Python, BASH, PowerShell, C/C++, Java, SystemVerilog, HTML/CSS/JS